AI SMART BLIND STICK

SDG AND TARGETS

- SDG 3: Good Health and Well-Being
  Target 3.6: Reduces accidents and injuries
- SDG 10: Reduced Inequalities
  Target 10.2: Promotes inclusion of visually impaired people
- SDG 11: Sustainable Cities and Communities
  Target 11.2: Ensures safe and accessible movement

PROBLEM STATEMENT

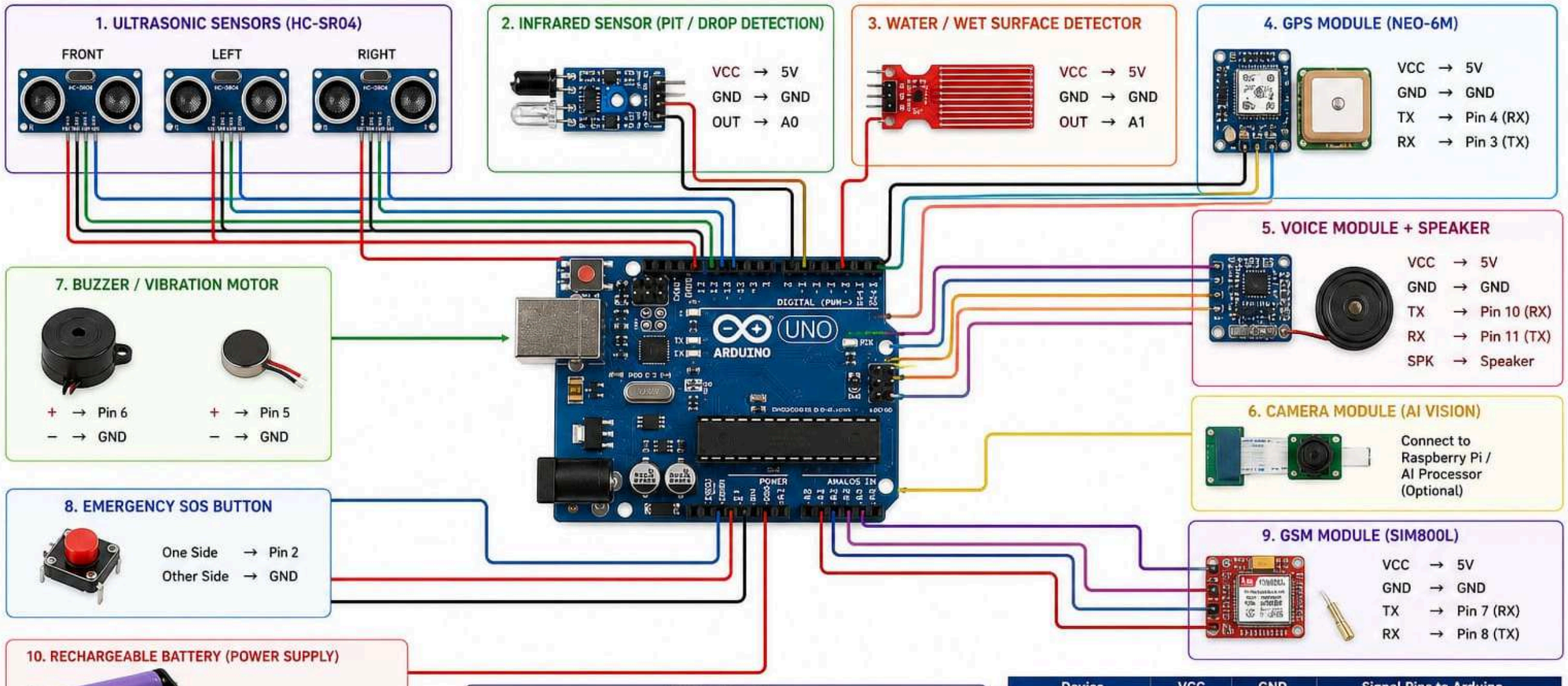
- Visually impaired people face many difficulties in their daily life while walking independently. They are unable to detect obstacles such as stones, pits, vehicles, and walls in their path. This increases the risk of accidents and injurie.
- They also face problems in identifying directions and crossing roads safely. Most of them depend on others for movement, which reduces their confidence and independence .
- Existing smart devices are expensive and not affordable for many people. There is a need for a low-cost, simple, and effective solution to help visually impaired people move safely and independently.

PROPOSED SOLUTION

- It detects nearby obstacles, pits, stairs, and moving objects and alerts the user through vibration and voice guidance.
- Ultrasonic Sensors (front, left, right): Detect obstacles up to 3 metres in all directions and trigger vibration/audio alerts .
- Infrared Sensor: Detects pits, open drains, stairs, and sudden drops in ground level.
- Water/Wet Surface Detector: Alerts users to slippery or flooded surfaces.
- GPS Module + Voice Navigation: Provides real-time location tracking and turn-by-turn voice guidance via earpiece
- Camera + AI Object Recognition: Uses a miniature AI vision module to identify people, vehicles, signage, and traffic signals
- Emergency SOS Button: One press sends GPS coordinates to a registered emergency contact via GSM .
- Rechargeable Battery: Up to 10 hours of operational use per charge via USB-C.

AI SMART BLIND STICK – COMPLETE CIRCUIT DIAGRAM

(With All Described Devices)



HOW IT WORKS

1. Ultrasonic sensors (Front, Left, Right) detect obstacles up to 3 metres.
2. IR sensor detects pits, stairs, or sudden drops.
3. Water sensor detects wet or slippery surfaces.
4. All sensor data is sent to Arduino.
5. If danger detected → Buzzer & Vibration ON + Voice alert.
6. GPS provides location & Voice module gives navigation.
7. SOS button → GSM sends location to emergency contact.
8. Camera + AI (optional) identifies objects, people, vehicles, signs.

Device	VCC	GND	Signal Pins to Arduino
Ultrasonic (Front)	5V	GND	Trig → D9, Echo → D10
Ultrasonic (Left)	5V	GND	Trig → D11, Echo → D12
Ultrasonic (Right)	5V	GND	Trig → D13, Echo → A2
IR Sensor	5V	GND	Out → A0
Water Sensor	5V	GND	Out → A1
Buzzer	5V	GND	+ → D6
Vibration Motor	5V	GND	+ → D5
GPS Module	5V	GND	TX → D4, RX → D3
Voice Module	5V	GND	TX → D10, RX → D11
SOS Button	5V	GND	One Side → D2
GSM Module	5V	GND	TX → D7, RX → D8

KEY FEATURES

Obstacle Detection | Pit / Drop Detection | Water Detection | Real-time GPS Navigation | AI Object Recognition | Voice Guidance | SOS Alert (GSM) | Long Battery Life (10 hrs)

SOFTWARE TOOLS

- Arduino IDE (C/C++) is used for programming the microcontroller (Arduino Uno/Nano) to process ultrasonic sensor data and control feedback mechanisms like buzzers.
- Audio Output (TTS): e-Speak or similar Text-to-Speech engines convert object data into voice alerts.
- Python combined with libraries such as OpenCV and models like YOLO (You Only Look Once) is used on Raspberry Pi to identify objects (people, cars, obstacles) in real-time via a camera module.
- YOLOX-Nano is used for efficient, edge-based object detection on Linux-enabled boards (e.g., Debian/Raspberry Pi).

PROJECT BUDGET

COMPONENT	QUANITITY	COAST
1. Ultrasonic Sensor(HC-SR04)	3	600
2. IR Sensor	1	150
3. Water Sensor	1	120
4. Arduino UNO	1	500
5. GPS Module(NEO-6M)	1	600
6. GSM Module(SIM800L)	1	500
7. Voice Module + Speaker	1	400
8. Camera Module(AI)	1	1500
9. Buzzer	1	50
10. Vibration Motor	1	100
11. Emergency Button	1	50
12. Rechargeable Battery	1	300
13. Stick(aluminium)	1	250

Total Coast: 5,320

IMPACT

- The AI Smart Blind Stick creates a positive impact on society by improving the lives of visually impaired people. It helps them move safely and independently without depending on others .
- The device reduces accidents by detecting obstacles, pits, and wet surfaces in advance. This increases user safety in daily activities like walking on roads and crossing streets.



CONCLUSION

The AI Smart Blind Stick is an assistive device that helps blind people walk safely using sensors and AI technology. It detects obstacles and alerts the user through sound or vibration, improving independence and quality of life.


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THE GLOBAL GOALS
For Sustainable Development